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ORIGINAL RESEARCH

STRUCTURAL

The Impact of Cerebral Embolic Protection Devices on Characteristics and Outcomes of Stroke Complicating TAVR

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ABSTRACT

BACKGROUND Acute ischemic stroke remains a serious complication of transcatheter aortic valve replacement (TAVR). Cerebral embolic protection devices (CEPD) were developed to mitigate the risk of acute ischemic stroke complicating TAVR (AISCT). However, the existing body of evidence does not clearly support CEPD efficacy in AISCT prevention.

OBJECTIVES In a cohort of patients with AISCT, we aimed to compare the characteristics and outcomes of patients who have had unprotected TAVR (CEPD−) vs CEPD-protected TAVR (CEPD+).

METHODS Data were derived from an international multicenter registry focusing on AISCT. We included all patients who experienced ischemic stroke within 72 hours of TAVR. Stroke severity was assessed using the National Institutes of Health Stroke Scale (NIHSS). Primary outcomes were neurologic disability status according to the modified Rankin Score at 30 days, and 6-month all-cause death. Propensity score matched analysis was used to control for differences between groups.

RESULTS In 18,725 TAVR procedures, 416 AISCT (2.2%) within 72 hours were documented, of which 376 were in the CEPD− TAVR group and 40 in the CEPD+ TAVR group. Although the middle cerebral artery stroke rate was similar in both groups (29.7% CEPD− vs 33.3% CEPD+; $P = 0.71$), AISCT in the CEPD+ group was characterized by a lower rate of internal carotid artery occlusion (0% vs 4.7%) and higher rate of vertebrobasilar system strokes (15.4% vs 5.7%; $P = 0.04$). AISCT was severe (NIHSS ≥15) in 21.6% CEPD− and 23.3% CEPD+ AISCT ($P = 0.20$). Disabling stroke rates (modified Rankin Score >1 at 30 days) were 47.3% vs 42.5% ($P = 0.62$), and 6-month mortality was 31.3% vs 23.3% ($P = 0.61$), in the CEPD− and CEPD+ groups, respectively. In the propensity score matched cohort, disabling stroke rates were 56.5% vs 41.6% ($P = 0.16$), and 6-month mortality was 33% vs 19.5% ($P = 0.35$), in the CEPD− and CEPD+ groups, respectively.

CONCLUSIONS In a large cohort of patients with AISCT, the use of CEPD had little effect on stroke distribution, severity, and outcomes. (J Am Coll Cardiol Intv 2024;17:666-677) © 2024 the American College of Cardiology Foundation. Published by Elsevier. All rights reserved.

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Transcatheter aortic valve replacement (TAVR) periprocedural complications, and particularly the occurrence of stroke have a marked impact on survival and quality of life.[1] In analyses of patients from the PARTNER (Placement of Aortic Transcatheter Valve) 2 studies and CENTER (Cerebrovascular EveNts in Patients Undergoing TranscathetER Aortic Valve Implantation With Balloon-expandable Valves Versus Self-expandable Valves) collaboration, major stroke after TAVR was associated with >5-fold increase in 1-year mortality.[1,2]

Despite the low rate of stroke complicating TAVR recorded in contemporary randomized controlled trials (RCTs) in low-risk patients,[3,4] the incidence of stroke after TAVR in a real-life population remains around 2% to 3%.[5] One-half of neurological events occur in the periprocedural period (~50% within 24 hours) and are thought to be related to cerebral embolization of debris or thrombus, and hypoperfusion.[6,7]

Cerebral embolic protection devices (CEPD) were developed to mitigate the risk of stroke related to debris embolization during TAVR. Thus far, 7 RCTs studying the effect of CEPD in TAVR have been published.[8-14] Pooled analysis of RCTs on CEPD showed an overall 30-day stroke incidence of 3.61% (95% CI: 2.57%-4.79%), with no significant reduction of ischemic lesions, 30-day stroke rates, or 30-day mortality in CEPD+ TAVR patients in comparison to CEPD− TAVR patients.[15] By contrast, observational studies show mixed results.[16-20] The most recent meta-analysis published included 5 RCTs and 5 inverse probability weighting/overall survival-adjusted observational studies, and showed no difference in 30-day stroke.[21]

The absence of proven efficacy in stroke prevention of CEPD raises questions regarding the mechanism of stroke in CEPD-protected TAVR vs non-CEPD TAVR.

We previously studied the incidence and characteristics of acute ischemic stroke complicating TAVR (AISCT) and described the indications and outcomes of interventions to treat AISCT in a multicenter registry (ASTRO-TAVI [Acute STROke Complicating TAVI - Management and Outcomes] collaboration; NCT05241821).[22]

In the current analysis, we aimed to study characteristics and outcomes of acute stroke complicating TAVR in CEPD-protected TAVR vs nonprotected TAVR in the ASTRO-TAVI registry.

METHODS

Data were derived from a multicenter registry focusing on TAVR patients complicated by periprocedural neurological events (the ASTRO-TAVI collaboration). Fourteen centers from Israel, Europe, and North America participated, providing data on consecutive cases of periprocedural stroke after TAVR. Data regarding clinical characteristics, case management, imaging, treatment, and outcomes were collected with a special focus on CEPD application. The registry protocol was approved by the local institutional review boards as required at each participating center and listed in ClinicalTrials.gov (NCT05241821). The Valve Academic Research Consortium 3 (VARC-3) definitions were used to define and categorize neurologic events.[23] Ischemic stroke was defined as the acute onset of focal neurological signs or symptoms conforming to a focal or multifocal vascular territory within the brain, spinal cord, or retina (NeuroARC Type 1a or 1aH) and fulfilling 1 of the following criteria: 1) signs or symptoms lasting ≥24 hours or until death, with pathology or neuroimaging evidence of central nervous system infarction, or absence of other apparent causes; or 2) symptoms lasting <24 hours, with pathology or neuroimaging confirmation of central nervous system infarction in the corresponding vascular territory.

ABBREVIATIONS AND ACRONYMS

AISCT = acute ischemic stroke complicating transcatheter aortic valve replacement

AV = aortic valve

CEPD = cerebral embolic protection device

MCA = middle cerebral artery

mRS = modified Rankin Score

NIHSS = National Institutes of Health Stroke Scale

PSM = propensity score matched

RCT = randomized controlled trial

TAVR = transcatheter aortic valve replacement

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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TABLE 1 Baseline Characteristics of Patients With Stroke After TAVR (CEPD− Compared with CEPD+)

	All Stroke (N = 416)	CEPD− (n = 376)	CEPD+ (n = 40)	*P* Value
Age, y	82.0 ± 7.4	82.1 ± 7.6	81.2 ± 4.6	0.28
Baseline NYHA functional class	2.9 ± 0.7	2.9 ± 0.7	2.7 ± 0.6	0.03
Creatinine, g/dL	1.3 ± 0.9	1.3 ± 1.0	1.1 ± 0.5	0.10
EuroSCORE II	4.0 (2.3-6.7)	4.2 (2.4-6.8)	2.9 (2.0-5.1)	0.02
Baseline mean gradient	42.5 ± 16.7	41.8 ± 16.0	49.3 ± 21.1	0.04
Valve area. cm^2	0.7 ± 0.2	0.7 ± 0.2	0.8 ± 0.2	0.19
Calcium score[a]	1,724 ± 1,915	1,494 ± 1,679	3,871 ± 2,621	<0.001
Female	232 (55.8)	214 (56.9)	18 (45.0)	0.18
Previous CABG	54 (13)	49 (13.0)	5 (12.5)	1.00
Previous PCI	139 (33.4)	129 (34.3)	10 (25.0)	0.06
Previous MI	69 (16.6)	64 (17.0)	5 (12.5)	0.65
Previous stroke	81 (19.5)	70 (18.6)	11 (27.5)	0.21
Previous valve surgery	17 (4.1)	15 (4.0)	2 (5.0)	0.67
Atrial fibrillation	149 (35.8)	137 (36.4)	12 (30.0)	0.74
Pacemaker	35 (8.4)	31 (8.2)	4 (10.0)	0.59
COPD	64 (15.4)	61 (16.2)	3 (7.5)	0.17
Diabetes mellitus	146 (35.1)	130 (34.6)	16 (40.0)	0.35
Hypertension	312 (75.0)	286 (76.1)	26 (65.0)	0.13
Bicuspid aortic valve	32 (7.7)	24 (6.4)	8 (20.0)	0.01
Valve in valve	12 (2.9)	10 (2.7)	2 (5.0)	0.32
General anesthesia[b]	102 (31.4)	97 (32.9)	5 (16.7)	0.10
Self expandable device[c]	265 (65.1)	237 (64.6)	28 (70.0)	0.60
Non-femoral access	75 (18.0)	72 (19.1)	3 (7.5)	0.08

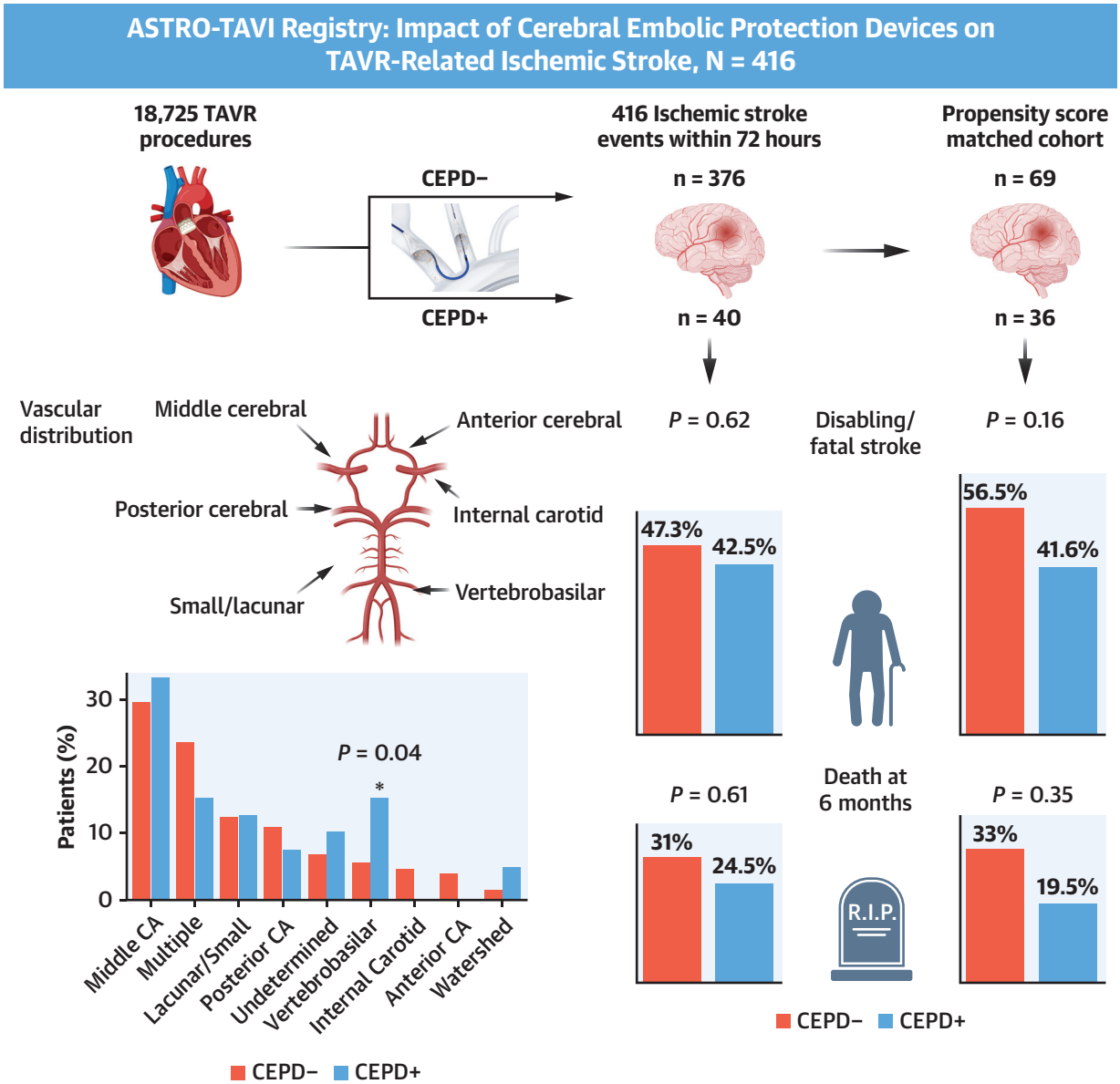
Values are mean ± SD, median (Q1-Q3), and n (%). [a]240 missing. [b]91 missing. [c]9 missing.

CABG = coronary artery bypass grafting; CEPD = cerebral embolic protection device; COPD = chronic obstructive pulmonary disease; MI = myocardial infarction; PCI = percutaneous coronary intervention; TAVR = transcatheter aortic valve replacement.

Stroke severity was assessed using the National Institutes of Health Stroke Scale (NIHSS), with an NIHSS of 0 to 5 considered to be a mild stroke, 6 to 14 moderate, and ≥15 severe.[24] Primary outcome measures were 6-month all-cause death, and neurologic disability status at 30 days (or at the latest follow-up available) according to the modified Rankin Score (mRS). Stroke disability status was classified as deceased (mRS 6), alive but dependent (mRS 3-5 and an increase of at least 1 from baseline), or alive and independent (mRS <3 or without an increase from baseline).[23] In case mRS or NIHSS records were unavailable in the patients' charts, centers were instructed to provide a categorical assessment of stroke disability (independent, dependent, deceased) and severity (mild, moderate, or severe) based on available medical records. Data regarding stroke severity, disability, and vascular territory localization were collected retrospectively based on neurological examination and neuroimaging documented in patients' charts, and were reviewed and adjudicated with neurologists when necessary.

To assess the impact of CEPD on the characteristics and outcomes of stroke complicating TAVR, we divided patients with AISCT within 72 hours of the procedure into 2 groups: CEPD-protected TAVR (CEPD+) and unprotected TAVR (CEPD−). Study endpoints were death, recurrent stroke, and recurrent rehospitalization at 6 months, stroke severity (NIHSS score), and stroke disability (mRS score at 30 days). Baseline characteristics were compared between the CEPD− and CEPD+ groups. Continuous variables are reported as mean ± SD or median (Q1-Q3), as appropriate. Categorical variables are described as percentages. Characteristics of study participants were compared using Student's *t*-test or Wilcoxon rank tests, as appropriate for continuous variables and chi-square test for categorical variables. All tests were 2-sided, and a *P* value <0.05 was considered significant. Death rates were graphically displayed using Kaplan-Meier curves and compared using the log rank test (unadjusted analysis). Logistic regression models were used to identify predictors of severe stroke, disabling stroke, and mortality at 6 months. Finally,

CENTRAL ILLUSTRATION **Study Flow Chart and Key Findings**



- Middle cerebral artery stroke rate did not significantly differ between groups
- A higher rate of vertebrobasilar stroke was observed in the CEPD+ group
- Disabling/fatal stroke and 6-month mortality did not significantly differ between groups

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ASTRO-TAVI = Acute STROke Complicating TAVI - Management and Outcomes; CA = cerebral artery; CEPD = cerebral embolic protection device; TAVR = transcatheter aortic valve replacement.

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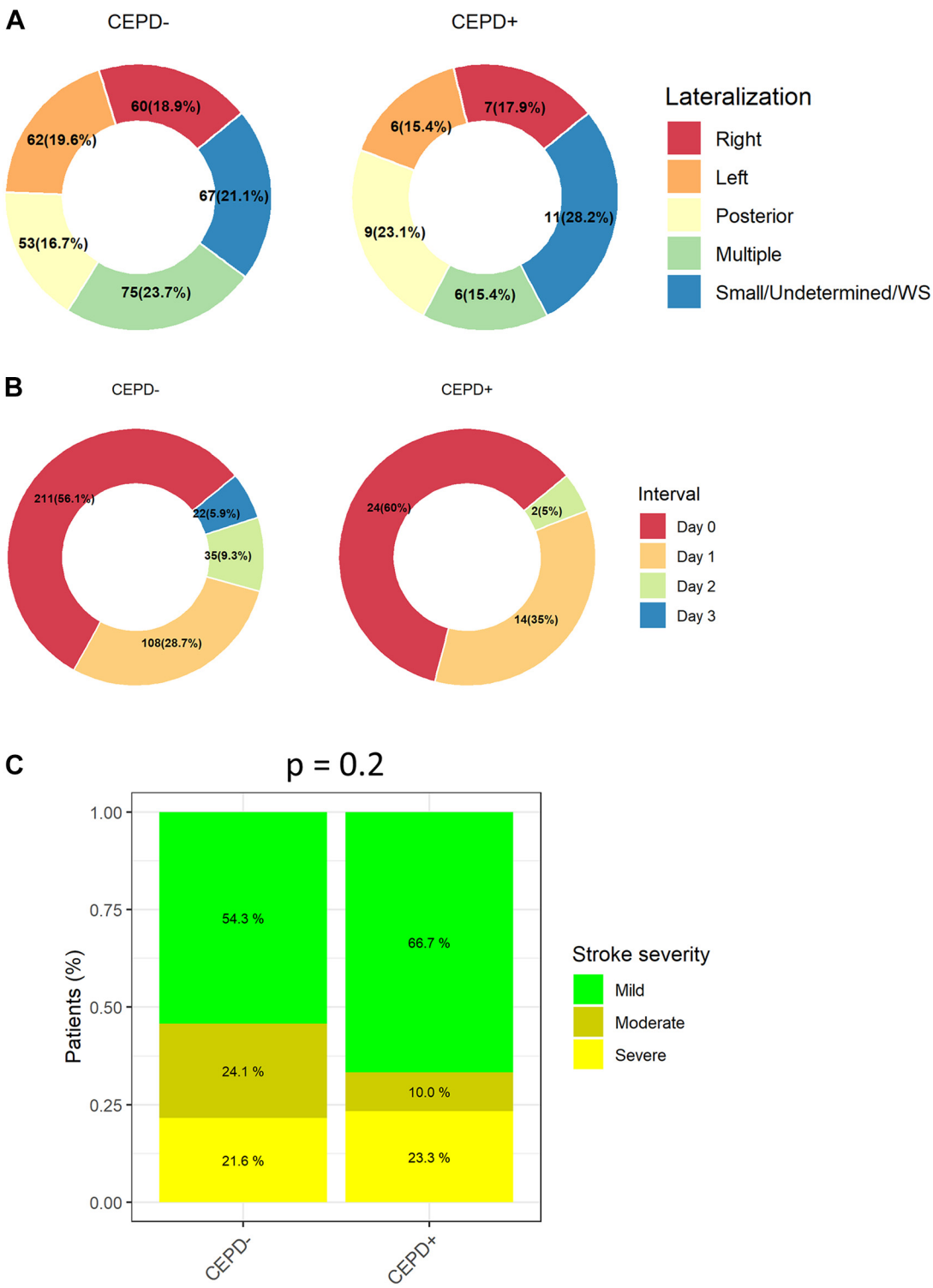
FIGURE 1 Stroke Lateralization, Interval From TAVR, and Severity Distribution



(A) Stroke lateralization: right, left, posterior circulation, multiple loci, or other (ie, lacunar stroke, undetermined, watershed); asterisks represents statistical significance; (B) temporal distribution of stroke in the cerebral embolic protection device (CEPD)− and CEPD+ groups; (C) National Institutes of Health Stroke Scale stroke severity in the CEPD− and CEPD+ groups.

TABLE 2 Stroke Characteristics of All Patients With Ischemic Stroke, CEPD− vs CEPD+

	All Stroke	CEPD−	CEPD+	*P* Value
Stroke severity	312	282	30	
Mild stroke: NIHSS 1-5	173 (55.4)	153 (54.3)	20 (66.7)	0.20
Moderate stroke: NIHSS 6-14	71 (22.8)	68 (24.1)	3 (10.0)	
Severe stroke: NIHSS ≥15	68 (21.8)	61 (21.6)	7 (23.3)	
Stroke territory	356	317	39	
Internal carotid	15 (4.2)	15 (4.7)	0 (0.0)	NA
Vertebrobasilar	24 (6.7)	18 (5.7)	6 (15.4)	0.04
Anterior CA	13 (3.7)	13 (4.1)	0 (0.0)	NA
Middle CA	107 (30.1)	94 (29.7)	13 (33.3)	0.71
Posterior CA	38 (10.7)	35 (11.0)	3 (7.7)	0.78
Lacunar/small	45 (12.6)	40 (12.6)	5 (12.8)	1.00
Watershed	7 (2.0)	5 (1.6)	2 (5.1)	0.17
Multiple	81 (22.8)	75 (23.7)	6 (15.4)	0.31
Undetermined	26 (7.3)	22 (6.9)	4 (10.3)	0.51

Values are n or n (%).
CA = cerebral artery; CEPD = cerebral embolic protection device; NIHSS = National Institutes of Health Stroke Scale.

propensity score matched (PSM) analysis was performed: We initially computed the propensity scores, which represent the probability of CEPD application given the observed covariates. These scores were derived using logistic regression, incorporating variables that may influence both the treatment assignment and the outcome of interest (EuroSCORE II, mean aortic valve [AV] gradient, bicuspid AV, nonfemoral access, late vs early procedure date). Subsequently, a nearest-neighbor 2:1 matching with a 0.1 caliper was employed to create a pseudo-population with well-balanced covariates across treatment groups. Outcomes were than compared between the groups. All analyses were performed using R V.4.0.0 software (R-studio).

RESULTS

Of 18,725 TAVR interventions performed in 14 centers between 2006 and 2022, AISCT within 72 hours occurred in 416 patients (2.22%). Median follow-up duration was 141 (Q1-Q3: 28-540) days. Data regarding CEPD utilization in the entire institutional TAVR cohort were available for 11 centers (n = 1,937/15,983; mean 12.1%, range 0% to 44.9%). In these centers, AISCT within 72 hours occurred in 2.4% of CEPD− TAVR procedures and in 2% of CEPD+ procedures ($P = 0.23$). Disabling/fatal stroke occurred in 1.1% of CEPD− and 0.9% CEPD+ AISCT ($P = 0.33$). The final cohort included 376 AISCT events occurring after CEPD− (TAVR performed between November 2006 and November 2022, median date January 18, 2017) and 40 after CEPD+ TAVR (TAVR performed between June 2012 and September 2022, median date March 4, 2020). All CEPD used in our cohort were SENTINEL systems (Boston Scientific). All CEPD implantations were proclaimed successful excluding 1. Baseline characteristics are provided in **Table 1**. Groups did not significantly differ in terms of age and sex distribution. Compared with CEPD− patients, CEPD+ patients had a significantly lower EuroSCORE II ($P = 0.02$), lower NYHA functional class ($P = 0.03$), and a higher mean baseline gradient ($P = 0.04$). Calcium score and bicuspid AV prevalence were higher in the CEPD+ group ($P < 0.001$ for both).

VESSEL DISTRIBUTION, SEVERITY, AND OUTCOMES.

Stroke vessel distribution, lateralization, and temporal distribution are shown in the **Central Illustration** and in **Figure 1**. Data regarding stroke vascular distribution were available in 356 of 416 patients (**Table 2**). Internal carotid occlusion was recorded in 15 (4.7%) of the CEPD− TAVR vs none in the CEPD+ TAVR. Stroke involving the middle cerebral artery (MCA) occurred in 94 (29.7%) of CEPD− and 13 (33.3%) of CEPD+ groups ($P = 0.71$). Stroke involving the vertebrobasilar artery occurred in 18 (5.7%) of the CEPD− group and 6 (15.4%) of the CEPD+ groups ($P = 0.04$). There was no statistically significant difference in stroke lateralization (classified as right-sided, left-sided, posterior circulation, multiple locations, or other [undetermined, lacunar/small, watershed]) (**Figure 1A**) or temporal distribution between groups (day 0, day 1, or day 2) (**Figure 1B**). In patients with documented stroke severity (75%), severity was mild in 54.3% and 66.7%, moderate in 24.1% and 10.0%, and severe in 21.6% and 23.3% in CEPD− and CEPD+ groups, respectively ($P = 0.20$) (**Figure 1C**). There was no difference in procedural

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TABLE 3 Outcomes: CEPD− vs CEPD+

	CEPD− (n = 376)	CEPD+ (n = 40)	*P* Value
Vascular complications	72 (19.4)	9 (22.5)	0.79
Minor	46 (12.4)	5 (12.5)	1.00
Major	26 (7.0)	4 (10.0)	0.707
Bleeding complications	91 (24.5)	10 (25.0)	1.00
Minor	29 (7.8)	2 (5.0)	0.748
Major	32 (8.6)	5 (12.5)	0.597
Life threatening	29 (7.8)	3 (7.5)	1.00
Acute kidney injury	87 (23.5)	6 (15.0)	0.306
Stage 1	54 (14.6)	3 (7.5)	0.321
Stage 2	13 (3.5)	1 (2.5)	1.00
Stage 3	19 (5.1)	2 (5.0)	1.00
Annular rupture	0 (0.0)	1 (2.5)	0.171
Tamponade	6 (1.6)	1 (2.5)	1.00
Coronary occlusion	3 (0.8)	0 (0.0)	NA
Successful implant	302 (84.8)	34 (87.2)	0.878
Acute myocardial infarction	4 (1.1)	0 (0.0)	NA
Need for pacemaker	64 (17.1)	3 (7.5)	0.181
In hospital death	83 (22.1)	8 (20.0)	0.913
Cardiac readmission	27 (16.4)	2 (8.0)	0.432
Intra cranial hemorrhage	7 (2.3)	1 (2.6)	1.00
Stroke post-discharge	34 (10.7)	2 (5.9)	0.965
Stroke disability			
No disability, mRS 0-1	171 (45.5)	22 (55.0)	0.62
With disability, mRS 2-5	93 (24.7)	8 (20.0)	
Death resulting from stroke	85 (22.6)	9 (22.5)	
Alive, mRS unknown	27 (7.2)	1 (2.5)	

Values are n (%).
CEPD = cerebral embolic protection device; mRS = modified Rankin Score.

complications between the groups (**Table 3**). Stroke disability distribution is shown in **Figure 2B**. Rates of disabling or fatal stroke (mRS >1) were 47.3% and 42.5% in the CEPD− and CEPD+ groups, respectively ($P = 0.62$). The Kaplan-Meier estimated death rate at 6 months was 31.3% vs 23.3% in the CEPD− and CEPD+ groups, respectively (log rank $P = 0.61$) (**Figure 2A**).

PREDICTORS OF STROKE SEVERITY, DISABILITY, AND DEATH. CEPD was not associated with a lower risk of severe stroke, neither in a univariate logistic regression model (OR: 1.1; 95% CI: 0.42-2.57; $P = 0.83$) (**Figure 3A**) nor in a multivariate model adjusted for EuroSCORE II and mean AV gradient (OR: 0.79; 95% CI: 0.25-2.01; $P = 0.63$). In terms of vascular territory, internal carotid, multiple loci, and MCA were associated with the highest risk of severe stroke (Supplemental Figure 1).

CEPD use was not associated with a lower rate of 6-month death following AISCT, neither in a univariate logistic regression model (OR: 0.6; 95% CI: 0.26-1.24; $P = 0.19$) (**Figure 3B**), nor in a multivariate model adjusted for EuroSCORE II and mean AV gradient (OR: 0.64; 95% CI: 0.26-1.4; $P = 0.28$). Several factors were identified as predictors of death after AISCT, most notably severe stroke ($P < 0.001$), but also male sex ($P = 0.004$), higher EuroSCORE II ($P = 0.007$), history of atrial fibrillation ($P = 0.004$), nonfemoral access ($P = 0.026$), general anesthesia ($P = 0.007$), and procedure date later than the median date (July 11, 2017; $P = 0.019$).

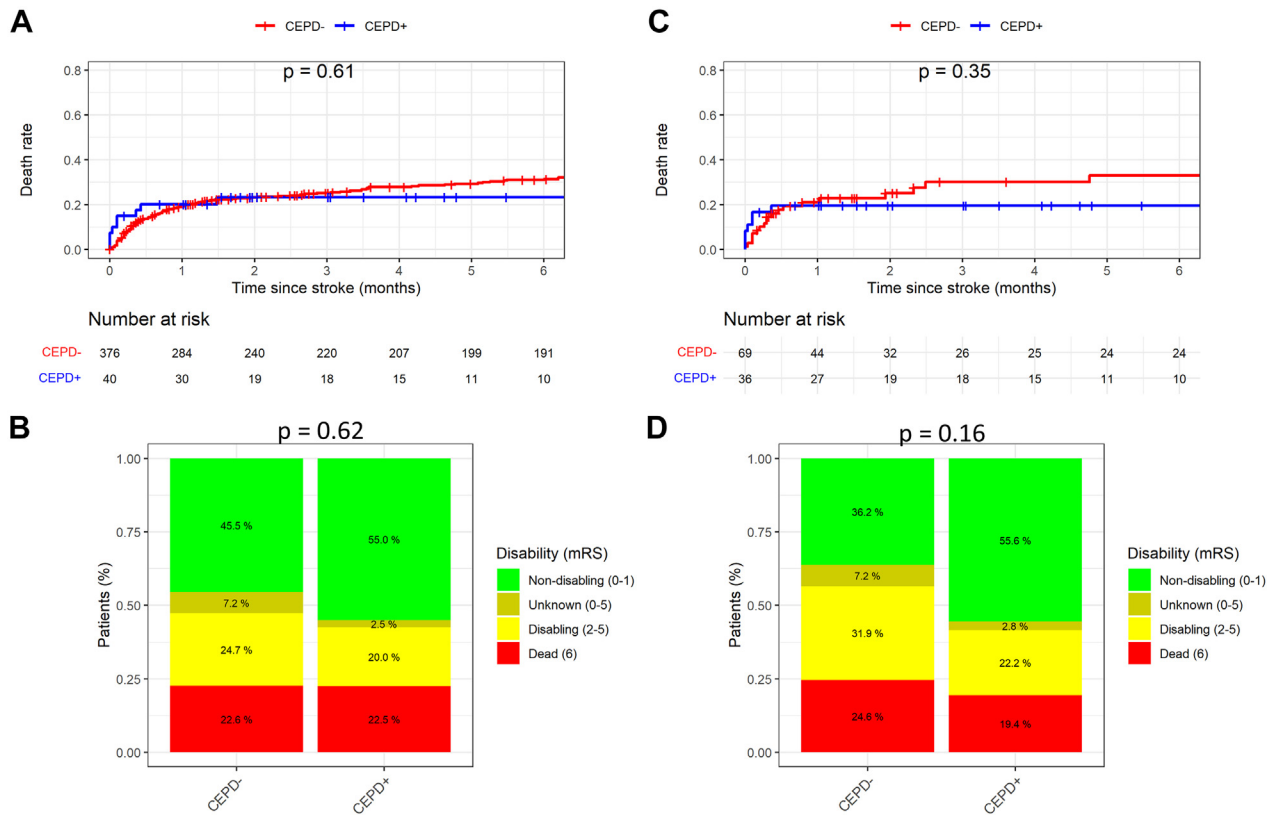
CEPD use was not associated with a lower rate of disabling or fatal stroke, neither in a univariate logistic regression model ($P = 0.56$) (**Figure 3C**) nor in a multivariate model adjusted for EuroSCORE II and mean AV gradient ($P = 0.32$). Compared with mild stroke, moderate stroke and severe stroke ($P < 0.001$ for both) were related to higher rates of disabling stroke. Male sex was associated with an increased risk of disabling or fatal stroke ($P = 0.02$). AISCT involving the MCA territory, internal carotid, and multiple locations were associated with the highest rate of disabling stroke (Supplemental Figure 2).

PROPENSITY SCORE MATCHING. In a 2:1 nearest-neighbor propensity score matching, 69 CEPD− patients were matched to 36 CEPD+ patients. Groups were adequately matched, with a standard mean difference <0.1 for all covariates. Propensity score distribution plots and mean standardized differences plots for all covariates included in the model, as well as descriptive tables of the PSM cohort, are provided in the Supplemental Figures 3 and 4 and Supplemental Tables 1 to 4. In the PSM cohort, rates of disabling or fatal stroke (mRS >1) were 56.5% and 41.6% in the CEPD− and CEPD+ groups, respectively ($P = 0.16$). Kaplan-Meier estimated death rate at 6 months 33.0% vs 19.5%, in the CEPD− and CEPD+ groups, respectively (log rank $P = 0.35$) (**Figure 2C**). In a logistic model adjusting for PSM, CEPD was not identified as a predictor of severe stroke, death at 6 months, or disabling stroke (Supplemental Figure 5).

DISCUSSION

In the current analysis from the ASTRO-TAVI collaboration, we have compared AISCT in CEPD-protected TAVR and unprotected TAVR. Our major findings are as follows: 1) the rate of AISCT at 72 hours was 2.2%, and the mean rate of CEPD use was ~12%; 2) CEPD use had no significant effect on the all stroke rate, disabling stroke rate (absolute risk reduction 0.2%; 95% CI: −0.45 to +0.86%), or on survival at 6 months; 3) within the stroke cohort, CEPD use did not significantly impact AISCT severity (severe stroke rate 21.6% and 23.3% in the CEPD− and CEPD+ groups,

FIGURE 2 Death Curves and Disability Distribution in the Entire Cohort and in the Propensity Score Matched Cohort



(A) Kaplan-Meier curve of 6-month survival in the cerebral embolic protection device (CEPD)− vs CEPD+ groups; (B) stroke disability at 30 days; (C) Kaplan-Meier curve of 6-month survival in the propensity score matched cohort; (D) stroke disability at 30 days in the propensity score matched cohort. mRS = modified Rankin Score.

respectively; $P = 0.20$); 4) although a higher rate of vertebrobasilar stroke and a lower rate of internal carotid artery strokes were recorded in the CEPD+ group compared with the CEPD− group, the most common arterial territory involved in both groups was the MCA; and 5) rates of disabling or fatal stroke (as assessed at 30 days) were 47.3% and 42.5% in the CEPD− and CEPD+ groups, respectively ($P = 0.62$). Differences remained nonsignificant in the PSM cohort.

The concept of CEPD to prevent stroke occurring during TAVR has gained popularity in recent years.[19,25] Initial experience with the SENTINEL device suggested significant debris entrapment within the device basket, which served to promote the use of this device.[26] However, robust evidence of clinical and imaging-verified stroke reduction failed to follow. Despite being conceptually promising, several mechanisms could potentially explain CEPD failure to prevent stroke. These include pore size that is larger than debris particles, insufficient contact with vessel wall, which may permit peridevice flow, microtrauma to the carotid/innominate wall during CEPD manipulation that could lead to stroke events, incomplete coverage of posterior circulation (unprotected left vertebral artery), other stroke mechanisms such as in situ thrombus formation, watershed strokes resulting from hemodynamic shift during TAVR in the context of fixed cerebral vascular stenosis, debris detachment occurring after CEPD removal, and left atrial appendage thrombus embolization.

In the PROTECTED TAVR (Stroke PROTECTion With SEntinel During Transcatheter Aortic Valve Replacement), the largest RCT published yet, 3,000 patients undergoing TAVR were assigned to receive cerebral protection or no cerebral protection.[14] Stroke events occurred in 34 (2.3%) and 43 (2.9%) in the CEPD and control arms, respectively ($P = 0.30$). The study failed to meet significance for the primary endpoint, and the incidence of clinical strokes within 72 hours or before discharge was similar across groups. Nevertheless, disabling stroke, defined as mRS score ≥2 at 1 month, was more

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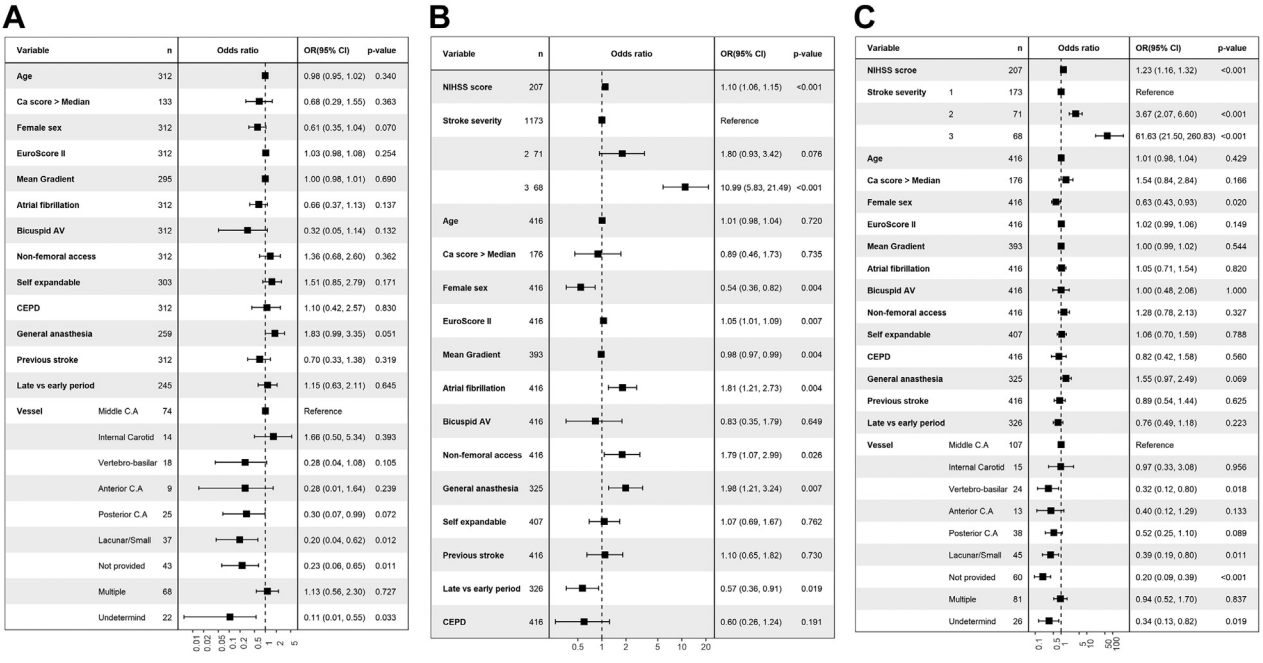
FIGURE 3 Logistic Regression Models Predicting Severe Stroke, Death, and Disabling/Fatal Stroke



Forest plots representing: (A) coefficients for univariate logistic regression with severe stroke as the outcome variable; (B) coefficients for univariate logistic regression with 6-month death the outcome variable; (C) coefficients for univariate logistic regression with disabling/fatal stroke as the outcome variable. AV = aortic valve; CA = cerebral artery; Ca score = calcium score; CEPD = cerebral embolic protection device; NIHSS = National Institutes of Health Stroke Scale.

prevalent in the control arm (n = 20, 1.33%) as compared with the CEPD arm (n = 8, 0.53%), suggesting a possible link between CEPD and reduction in stroke severity, rather than in stroke incidence. Following this logic, and based on a 0.8% reduction in disabling stroke, 125 CEPD would have to be placed to prevent 1 disabling stroke. We intended to test this hypothesis by comparing AISCT occurring in CEPD-protected and unprotected TAVRs. Stroke rates in our cohort were in keeping with previous registry-based reports[5] and slightly lower than the reported rate in the PROTECTED TAVR trial (2.2% vs 2.6%, respectively). This difference could be explained by under-reporting of mild strokes in the real-life scenario resulting from the absence of a routine neurologic exam.[27]

Some differences in baseline characteristics between the groups were observed and can be roughly categorized into 2 groups: 1) A lower median EuroSCORE II, lower mean NYHA functional class score, higher mean AV gradient, and a numerically lower prevalence of both nonfemoral access and general anesthesia in the CEPD+ compared with the CEPD− group probably reflect a shift toward selection of lower-risk aortic stenosis patients for TAVR, combined with the adoption of a more minimalistic approach to TAVR in the contemporary practice (given the later median date of CEPD+ TAVR procedures); and 2) A higher calcium score, higher prevalence of bicuspid AV, and a numerically higher previous stroke rate in the CEPD+ compared with the CEPD− group may reflect a predilection toward a selective application of CEPD devices in patients that were perceived to be at high risk of stroke.[2,28-30] In the current analysis, neither bicuspid AV, higher calcium score, nor previous stroke were associated with stroke severity, disability, or death at 6 months. Although gender was not previously identified as a determent of AISCT,[31] our analysis suggests improved outcomes after AISCT in women as compared with men.

Considering vascular distribution, it seems that CEPD was effective in preventing internal carotid strokes and strokes of multiple territories, and ineffective in reducing vertebrobasilar strokes. MCA was the most common vascular territory, occurring in around one-third of all patients. Surprisingly, CEPD use did not reduce the MCA stroke rate. In keeping with our findings, a single-center analysis by Isogai et al[32] reported that 5 of 8 ischemic strokes in

CEPD-protected TAVR involved the MCA. Notwithstanding, in a PSM study from Rotterdam and Groningen, at 24 hours, 2 of 3 neurological events (67%) in the CEPD group were located in protected areas vs 10 of 12 (83%) in the non-CEPD group.[20]

Given that dual-basket CEPD are intended to protect both carotid arteries and thus prevent MCA strokes, our finding emphasizes both the shortcomings of this system and our incomplete understating of AISCT mechanism and timing.[33,34] Nevertheless, there is a signal suggesting that CEPD were related to some shift in vascular territory distribution, reducing internal carotid artery strokes and increasing vertebrobasilar strokes.

Stroke severity was roughly similar between groups (21.6% and 23.3% severe stroke in the CEPD− and CEPD+ groups, respectively; $P = 0.20$). Previous data regarding comparative AISCT severity are limited. Isogai et al[32] reported nominally lower NIHSS in CEPD+ vs CEPD− AISCT; however, this was based on a single-center experience with an exceptionally low stroke rate and consequently a small number of events in both groups (15 and 8 in the CEPD− and CEPD+ groups, respectively). No information regarding stroke severity was reported for the stroke cohort in the PROTECTED TAVR.

Disabling or fatal stroke rates (mRS >1 as evaluated 30 days post-stroke) in the CEPD− group compared with the CEPD+ group were 47.3% and 42.5%, respectively, in the entire cohort ($P = 0.62$), and 56.5% and 41.6%, respectively, in in the PSM cohort ($P = 0.16$). By comparison, the disabling stroke rate in the PROTECTED TAVR trial was 46.5% (20/43 strokes) and 23.5% (8/34 strokes) as a fraction of total strokes in the CEPD− and CEPD+ groups, respectively, reflecting a higher stroke disability among CEPD+ within the stroke group ($P = 0.056$). In other words, in contrast to our observation, in the PROTECTED trial, although CEPD use did not reduce the overall stroke rate, it did reduce stoke complexity. Our outcomes are complementary with some of the previous publications, which did not identify disabling stroke reduction with CEPD,[10,12,26] and conflict with others.[14,16,20] Naturally, the small overall number of CEPD+ strokes subject our comparison to type II error. Nevertheless, given the denominator of more than 1,900 CEPD+ TAVR patients, it could be argued that even if CEPD mediated some reduction in disabling stroke rate, the effect size would be exceedingly small.

The major strength of the current analysis is the large number of AISCT analyzed. Given that only stroke cases were collected and reviewed, we were able to describe the largest number thus far of strokes occurring in CEPD-protected TAVR patients with high granularity.

STUDY LIMITATIONS. Data regarding stroke severity were lacking for 25% of patients. Data regarding stroke extent per imaging modalities such as computed tomography perfusion or diffusion weighted magnetic resonance imaging, which could have shed some more light on stroke extent, are lacking. All CEPD+ stroke cases occurred with the SENTINEL device, and conclusions should therefore not be extended to other CEPD systems. Finally, our data are registry-based, and are thus subjected to an inherent selection bias with patients perceived at higher risk of stoke selected to TAVR with CEPD. We addressed this issue by adjusting to the major differences in baseline characteristics using a multivariate model and by using a PSM analysis.

CONCLUSIONS

In our analysis of 416 patients with AISCT within 72 hours of TAVR, we found subtle differences in stroke territorial distribution, and no significant difference in stroke severity in CEPD− compared with CEPD+ patients. Accordingly, the disabling stroke rate was only slightly lower in CEPD-protected TAVR, not reaching statistical significance. The role of CEPD in the current TAVR practice remains controversial with several remaining ethical and scientific questions: Given the low stroke rates encountered, could larger CEPD trials demonstrate clinical stroke reduction? If so, what would be the threshold of number needed to treat to prevent 1 disabling stroke that would justify this intervention in terms of cost effectiveness? These questions will be addressed with the publication of the ongoing BHF PROTECT-TAVI trial (British Heart Foundation Randomised Trial of Routine Cerebral Embolic Protection in Transcatheter Aortic Valve Implantation), aiming to recruit 7,730 participants randomized in a 1:1 ratio to undergo TAVR with CEPD or without CEPD.[35] Another issue that is yet to be resolved is whether CEPD could prevent subclinical stroke and future cognitive deterioration. Finally, even with the most optimistic perspective regarding the future of TAVR and CEPD, periprocedural strokes are not entirely preventable. It is therefore crucial to develop a pathway for stroke identification and management as previously discussed by our group and others.[22,36]

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PERSPECTIVES

WHAT IS KNOWN? AIS is a feared complication of TAVR. The role of CEPD in the prevention of AIS complicating TAVR remains unclear.

WHAT IS NEW? In patients with AIS complicating TAVR, CEPD use did not significantly impact stroke distribution, lateralization, or TAVR to stroke interval. Accordingly, CEPD use had no significant effect on stroke disability and survival.

WHAT IS NEXT? It is yet to be determined in larger studies whether specific population subsets could derive benefit from CEPD use. Attention should also be given to effective stroke treatment.

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KEY WORDS cerebral embolic protection device, complications, registry, SENTINEL, transcatheter aortic valve replacement

APPENDIX For supplemental figures and tables, please see the online version of this paper.